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What are Foundation Models?

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What is a Foundation Model?



Trained on massive datasets, foundation models (FMs) are large [deep learning](#) neural networks that have changed the way data scientists approach [machine learning \(ML\)](#). Rather than develop [artificial intelligence \(AI\)](#) from scratch, data scientists use a foundation model as a starting point to develop ML models that power new applications more quickly and cost-effectively. The term *foundation model* was coined by researchers to describe ML models trained on a broad spectrum of generalized and unlabeled data and capable of performing a wide variety of general tasks such as understanding language, generating text and images, and conversing in natural language.

What is unique about foundation models?

A unique feature of foundation models is their ability to handle a wide range of disparate tasks with a high degree of accuracy. For example, foundation models can be used for natural language processing (NLP), image processing, and code generation. The nature of FMs make them different from traditional ML models, which typically perform specific tasks like analyzing text for sentiment.

You can use foundation model applications. These models are trained on massive datasets and are often larger in size and complexity.



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For example, BERT, one of the first bidirectional foundation models, was released in 2018. It was trained using 340 million parameters and a 16 GB training dataset. In 2023, only five years later, OpenAI trained GPT-4 using 170 trillion parameters and a 45 GB training dataset. According to OpenAI,



the [computational power required for foundation modeling has doubled](#) every 3.4 months since 2012. Today's FMs, such as the large language models (LLMs) Claude 2 and Llama 2, and the text-to-image model Stable Diffusion from Stability AI, can perform a range of tasks out of the box spanning multiple domains, like writing blog posts, generating images, solving math problems, engaging in dialog, and answering questions based on a document.

[Read about large language models »](#)

Why is foundation modeling important?

Foundation models are poised to significantly change the machine learning lifecycle. Although it currently costs millions of dollars to develop a foundation model from scratch, they're useful in the long run. It's faster and cheaper for data scientists to use pre-trained FMs to develop new ML applications rather than train unique ML models from the ground up.

One potential use is automating tasks and processes, especially those that require reasoning capabilities. Here are a few applications for foundation models:

- Customer support
- Language translation
- Content generation
- Copywriting
- Image classification
- High-resolution image creation and editing
- Document extraction
- Robotics
- Healthcare
- Autonomous vehicles



How do foundation models work?

Foundation models are a form of generative artificial intelligence (generative AI). They generate output from one or more inputs (prompts) in the form of human language instructions. Models are based on complex neural networks including generative adversarial networks (GANs), transformers, and variational encoders.

Although each type of network functions differently, the principles behind how they work are similar. In general, an FM uses learned patterns and relationships to predict the next item in a sequence. For example, with image generation, the model analyzes the image and creates a sharper, more clearly defined version of the image. Similarly, with text, the model predicts the next word in a string of text based on the previous words and its context. It then selects the next word using probability distribution techniques.

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Foundation models use self-supervised learning to create labels from input data. This means no one has instructed or trained the model with labeled training data sets. This feature separates LLMs from

previous ML architectures, which use supervised or unsupervised learning.

[Read about machine learning »](#)

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What can foundation models do?

Foundation models, even though are pre-trained, can continue to learn from data inputs or prompts during inference. This means that you can develop comprehensive outputs through carefully curated prompts. Tasks that FM can perform include language processing, visual comprehension, code generation, and human-centered engagement.

Language processing

These models have remarkable capabilities to answer natural language questions and even the ability to write short scripts or articles in response to prompts. They can also translate languages using NLP technologies.

Visual comprehension

FMs excel in computer vision, especially with regard to identifying images and physical objects. These capabilities may find use in applications such as autonomous driving and robotics. Another capability is the generation of images from input text, as well as photo and video editing. ☆

Code generation

Foundation models can generate computer code in various programming languages based on natural language inputs. It's also feasible to use FMs to evaluate and debug code. [Learn more about AI code generation.](#)

Human-centered engagement

Generative AI models use human inputs to learn and improve predictions. An important and sometimes overlooked application is the ability of these models to support human decision-making. Potential uses include clinical diagnoses, decision support systems, and analytics.

Another capability is the development of new AI applications by fine-tuning existing foundation models.

Speech to text

Since FMs understand language, they can be used for speech to text tasks such as transcription and video captioning in a variety of languages.

What are examples of foundation models?

The number and size of foundation models on the market have grown at a rapid pace. There are now dozens of models available. Here is a list of prominent foundation models released since 2018.

BERT

Released in 2018, Bidirectional Encoder Representations from Transformers (BERT) was one of the first foundation models. BERT is a bidirectional model that analyzes the context of a complete sequence then makes a prediction. It was trained on a plain text corpus and Wikipedia using 3.3 billion tokens (words) and 340 million parameters. BERT can answer questions, predict sentences, and translate texts.

GPT

The Generative Pre-trained Transformer (GPT) model was developed by OpenAI in 2018. It uses a 12-layer transformer decoder with a self-attention mechanism. And it was trained on the BookCorpus dataset, which holds over 11,000 free novels. A notable feature of GPT-1 is the ability to do zero-shot learning.

GPT-2 released in 2019. OpenAI trained it using 1.5 billion parameters (compared to the 117 million parameters used on GPT-1). GPT-3 has a 96-layer neural network and 175 billion parameters and is trained using the 500-billion-word Common Crawl dataset. The popular ChatGPT [chatbot](#) is based on GPT-3.5. And GPT-4, the latest version, launched in late 2022 and successfully passed the Uniform Bar Examination with a score of 297 (76%).

[Read more about GPT »](#)

Amazon Nova

[Amazon Nova](#) is a new generation of state-of-the-art (SOTA) foundation models (FMs) that deliver frontier intelligence and industry leading price-performance.



- Amazon Nova Micro, Amazon Nova Lite, and Amazon Nova Pro are understanding models that accept text, image, and video inputs and generate text output. They provide a broad selection of capability, accuracy, speed, and cost operation points. Learn more about [Amazon Nova understanding models](#).
- Amazon Nova Canvas and Amazon Nova Reel are creative content generation models that accept text and image inputs and produce image or video outputs. They are designed to deliver customizable high-quality images and videos for visual content generation. Learn more about [Amazon Nova creative content generation models](#).

AI21 Jurassic

Released in 2021, Jurassic-1 is a 76-layer auto-regressive language model with 178 billion parameters. Jurassic-1 generates human-like text and solves complex tasks. Its performance is comparable to GPT-3.

In March 2023, [AI21 Labs](#) released Jurassic-2, which has improved instruction following and language capabilities.

Claude

Claude 3.5 Sonnet

Anthropic's most intelligent and advanced model, [Claude 3.5 Sonnet](#), demonstrates exceptional capabilities across a diverse range of tasks and evaluations while also outperforming Claude 3 Opus.

Claude 3 Opus

Opus is a highly intelligent model with reliable performance on complex tasks. It can navigate open-ended prompts and sight-unseen scenarios with remarkable fluency and human-like understanding. Use Opus to automate tasks, and accelerate research and development across a diverse range of use cases and industries.

Claude 3 Haiku

Haiku is Anthropic's fastest, most compact model for near-instant responsiveness. Haiku is the best choice for building seamless AI experiences that mimic human interactions. Enterprises can use Haiku to moderate content, optimize inventory management, produce quick and accurate translations, summarize unstructured data, and more.

Cohere

[Cohere](#) has two LLMs: one is a generation model with similar capabilities as GPT-3 and the other is a representation model intended for understanding languages. While Cohere has only 52 billion parameters, it outperforms GPT-3 in many respects.

Stable Diffusion

[Stable Diffusion](#) is a text-to-image model that can generate realistic-looking, high-definition images. It



need an extensive computing infrastructure. Stable Diffusion will run on a normal graphics card or even on a smartphone with a Snapdragon Gen2 platform.

[Read more about Stable Diffusion »](#)

BLOOM

BLOOM is a multilingual model with similar architecture to GPT-3. It was developed in 2022 as a collaborative effort involving over a thousand scientists and the Hugging Space team. The model has 176 billion parameters and training took three and a half months using 384 Nvidia A100 GPUs. Although the BLOOM checkpoint requires 330 GB of storage, it will run on a standalone PC with 16 GB of RAM. BLOOM can create text in 46 languages and write code in 13 programming languages.

Hugging Face

Hugging Face is a platform that offers open-source tools for you to build and deploy machine learning models. It acts as a community hub, and developers can share and explore models and datasets. Membership for individuals is free, although paid subscriptions offer higher levels of access. You have public access to nearly 200,000 models and 30,000 datasets.

What are challenges with foundation models?

Foundation models can coherently respond to prompts on subjects they haven't been explicitly trained on. But they have certain weaknesses. Here are some of the challenges facing foundation models:

- **Infrastructure requirements.** Building a foundation model from scratch is expensive and requires enormous resources, and training may take months.
- **Front-end development.** For practical applications, developers need to integrate foundation models into a software stack, including tools for prompt engineering, fine-tuning, and pipeline engineering.
- **Lack of comprehension.** Although they can provide grammatically and factually correct answers, foundation models have difficulty comprehending the context of a prompt. And they aren't socially or psychologically aware.
- **Unreliable answers.** Answers to questions on certain subject matter may be unreliable and sometimes inappropriate, toxic, or incorrect.
- **Bias.** Bias is a distinct possibility as models can pick up hate speech and inappropriate undertones from training datasets. To avoid this, developers should carefully filter training data and encode specific norms into their models.

How can AWS help with your foundation model needs?

[Amazon Bedrock](#) is the easiest way to build and scale [generative AI](#) applications with foundation models. Amazon Bedrock is a fully managed service that makes foundation models from Amazon and leading AI startups available through an API, so you can choose from various FMs to find the model that's best suited for your use case. With Bedrock, you can speed up developing and deploying scalable, reliable, and secure generative AI applications without managing infrastructure. ☆

[Amazon SageMaker JumpStart](#), which is a ML hub offering models, algorithms, and solutions, provides access to hundreds of foundation models, including top performing publicly available foundation models. New foundation models continue to be added, including [Llama 2](#), [Falcon](#), and [Stable Diffusion XL 1.0](#).

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